

Zachary Phillips

Senior .NET Software Engineer | AI & ML

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Senior AI-Enabled Software Engineer with **9+ years of experience in .NET engineering and AI-driven systems**, delivering production-grade solutions across logistics, finance, and customer operations. Expert in **agentic AI platforms, RAG architectures, and ML-powered automation** using **Azure OpenAI, LangChain, Python, and .NET**, combining LLM reasoning, vector search, and event-driven workflows at scale. Strong background in **computer vision, OCR, and intelligent document processing**, driving measurable improvements in efficiency, accuracy, and regulatory compliance. Proven leader in **hybrid cloud-native architectures** leveraging microservices, Docker/Kubernetes, and Azure CI/CD, with end-to-end **full-stack ownership** including **React and Angular frontends**, API design, and enterprise system integrations.

EDUCATION

Degree of Computer Science

Aug 2011 – May 2015

Eastern University | St Davids, PA

EXPERIENCE

Senior AI-Enabled .NET Software Engineer

Feb 2022 – Dec 2025

Trigent Software | Minneapolis, MN

- Architected and delivered an **Agentic AI platform** using **Azure OpenAI (GPT-4/3.5)** and **LangChain**, implementing autonomous logistics, customer service, and finance agents with tool calling across **ERP systems, carrier REST/GraphQL APIs, and Twilio**, enabling adaptive workflow execution for customs exceptions, routing changes, and customer communications.
- Built a **Retrieval-Augmented Generation (RAG)** solution with **LangChain**, embedding models, and **vector databases (Azure Cognitive Search / FAISS)** to provide semantic search over shipping regulations and compliance documentation, allowing operations teams to query complex international requirements using natural language with source-grounded responses.
- Developed **AI-driven document processing pipelines** using **Python microservices, Azure Document Intelligence / Form Recognizer**, OCR, and document classification models to automatically identify customs declarations, VAT forms, and invoices, reducing manual review effort by nearly **64%** while improving data accuracy and regulatory consistency.
- Implemented **ML-based carrier selection and pricing optimization** using **Python (scikit-learn)**, combining shipment dimensions, weight, declared value, and historical carrier performance to generate automated cost vs. delivery-time recommendations and adaptive routing decisions beyond static rule-based systems.
- Engineered event-driven AI automation using **webhooks, message queues**, and modular API connectors, enabling agents to react to ERP updates in real time, regenerate shipping documents, trigger **Twilio SMS/email notifications**, and propose alternative delivery routes—reducing customer shipment inquiries by over **55%**.
- Led the platform's transition to a **hybrid AI architecture**, integrating **.NET and Python microservices, OpenAI GPT reasoning models, LangChain agent orchestration, PostgreSQL**, vector search, and **Docker/Kubernetes on Azure**, delivering scalable, resilient deployment of multi-agent systems across global logistics operations.
- Owned the full **Full-Stack development lifecycle**, from system design and API architecture to UI delivery - building **React + Tailwind** dashboards for real-time shipment tracking, implementing **REST & GraphQL** integrations with ERP and carrier platforms, containerizing services with **Docker**, orchestrating deployments via **Kubernetes**, and delivering cloud-native AI solutions through Agile CI/CD pipelines on **Azure**.

EXPERIENCE

Senior Software Engineer (.NET/Computer Vision)

Nov 2018 – Jan 2022

Emergent Software | Southborough, MA

- Architected and delivered an **AI-powered Computer Vision platform** using **Azure Cognitive Services Computer Vision API (v2.1-v3.1)**, implementing pipelines for **object detection, image classification, automated tagging, caption generation, and OCR**, enabling large-scale processing of unstructured visual data across multiple industries.
- Designed end-to-end **image analysis workflows** integrating **Azure Computer Vision REST APIs, OpenCV**, and custom **.NET Core 2.2 / 3.1 (ASP.NET Core Web APIs)** services to extract bounding boxes, confidence scores, object categories, and visual descriptors, supporting **content-based image retrieval (CBIR)** and ML dataset generation.
- Built AI-driven **visual data enrichment pipelines** combining pretrained Azure Vision models with custom post-processing logic to normalize coordinates, generate structured metadata, and cluster detected entities (persons, objects, animals), transforming raw images into ML-ready datasets for recommendation engines and intelligent search.
- Implemented **OCR and document digitization pipelines** using Azure Computer Vision Read API, enabling extraction of printed and handwritten text, layout-aware processing, and semantic indexing of scanned documents and mixed visual content.
- Engineered scalable backend services leveraging **Azure Blob Storage**, asynchronous processing, and REST-based orchestration to support batch uploads, parallel image inference, and persistent storage of extracted features for analytics and continuous optimization.
- Integrated computer vision outputs into higher-level **ML workflows**, enabling dataset curation, feature normalization, and preparation for downstream supervised learning and similarity matching systems.
- Developed interactive visualization and annotation tools using **Angular 8-10, Tailwind CSS, Fabric.js**, and HTML5 Canvas, rendering bounding boxes and overlays to support **human-in-the-loop validation**, confidence review, and assisted labeling.
- Delivered full-stack cloud solutions using **ASP.NET Core 2.2/3.1, Angular, Azure App Services, Azure Blob Storage**, and RESTful APIs, owning solution architecture, AI integration, frontend development, and Azure deployment pipelines.

.NET Software Engineer

Sep 2015 – Oct 2018

Saritas | Newport Beach, CA

- Modernized a legacy web-based bus ticketing system into a **cross-platform solution** (web, mobile, desktop) using **Blazor Hybrid (Blazor + .NET MAUI)**, achieving **90%+ code reuse** and consistent performance across devices.
- Designed a **modular, cloud-hosted architecture** on **Microsoft Azure**, enabling rapid feature expansion, simplified maintenance, and continuous platform evolution.
- Built scalable backend services with **ASP.NET Core, Entity Framework, RESTful APIs, GraphQL, gRPC**, and **WebSockets**, backed by **MS SQL Server** and **Azure PostgreSQL**, leveraging **Redis Cache** and **async processing** to reduce latency and sustain peak-load throughput.
- Implemented **payment and booking microservices** using **Kafka** and **RabbitMQ** for event-driven and message-based communication; integrated **Stripe** and developed a **custom bill-splitting algorithm** with shareable invoice links and real-time payment reconciliation.
- Deployed and operated containerized services using **Docker, Kubernetes**, and **Azure Container Apps**, ensuring reliable **CI/CD pipelines** and consistent runtime environments.
- Implemented end-to-end **security and observability**, including **OAuth 2.0, JWT, SSL/TLS, RBAC/ReBAC**, and centralized monitoring with **Azure Application Insights** and **Grafana**; integrated **Twilio** for real-time user notifications (booking status, payment confirmations), contributing to a **3x increase in user satisfaction**.

PROJECTS

Cross-Platform Bus Ticketing Platform

<https://tripXpress.com>

Modernized a legacy bus ticketing platform into a high-performance cross-platform solution spanning web, mobile, and desktop. By leveraging Blazor Hybrid and a unified .NET codebase, the project achieved over 90% code reuse, significantly reduced maintenance overhead, and delivered consistent performance across devices. The new Azure-hosted modular architecture enabled rapid feature expansion, reliable payments with real-time bill splitting, and improved system observability—resulting in a 3× increase in user satisfaction and a future-ready platform built for continuous growth.

Intelligence Shipping Orchestration Platform

<https://lelogistics.net>

Led the development of an AI-powered Intelligent Shipping Orchestration Platform for a global logistics enterprise, modernizing legacy systems into a cloud-native, agentic AI solution on Azure. Implemented RAG-based knowledge retrieval, autonomous multi-agent workflows, and ML-driven carrier optimization using Azure OpenAI, LangChain, and Python/.NET microservices—cutting manual document processing by 64%, reducing customer shipment inquiries by 55%, and enabling real-time, automated decision-making across routing, compliance, and customer communications.

PERSONAL SKILLS

Back-End & APIs:

.NET (ASP.NET Core 2.2–7), C#, Python, RESTful APIs, GraphQL, gRPC, WebSockets, Entity Framework, Asynchronous & Event-Driven Processing, Webhooks

AI & Machine Learning:

Agentic AI Systems, Autonomous Agents, LLM Integration (Azure OpenAI GPT-4 / GPT-3.5), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Embeddings, Semantic Search, Vector Databases (Azure Cognitive Search, FAISS), scikit-learn, Feature Engineering, Model Evaluation

Strong Communication:

Azure Cognitive Services Computer Vision (v2.1–v3.1), Azure Document Intelligence / Form Recognizer, OpenCV, Object Detection, Image Classification, OCR (Printed & Handwritten), Document Classification, Image Annotation, Bounding Boxes, Confidence Scoring

Front-End:

React, Angular (8–10), Blazor Hybrid (.NET MAUI), TypeScript, JavaScript, HTML5, CSS3, Bootstrap, Tailwind CSS, Fabric.js, Canvas-based Visualization, Real-Time Dashboards

Cloud, DevOps & Infrastructure:

Microsoft Azure (Azure OpenAI, App Services, Blob Storage, Azure Container Apps), Docker, Kubernetes, CI/CD (Azure DevOps, GitHub Actions), PostgreSQL, MS SQL Server, Redis Cache

Messaging, Streaming & Integrations:

Kafka, RabbitMQ, ERP Integrations, Carrier APIs (REST / GraphQL), Twilio (SMS, Email, Notifications), Stripe Payments

Security, Observability & Reliability:

OAuth 2.0, JWT, SSL/TLS, RBAC / ReBAC, Azure Application Insights, Grafana, Logging, Monitoring, Performance Optimization

Architecture & Engineering Practices:

Microservices Architecture, Hybrid AI Systems (.NET + Python), Cloud-Native Design, System Architecture, Agile / Scrum, Full SDLC Ownership